

Liposomal Bupivacaine-Based Regional Anesthesia Provides Excellent Pain Control and Minimizes Opioid Consumption Following Anterior Cruciate Ligament Reconstruction: A Prospective Randomized Controlled Trial

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Objectives:

Femoral nerve blockade (FNB), adductor canal blocks (ACB) and the interspace between the popliteal artery and capsule of the posterior knee (IPACK) blocks are common techniques for regional analgesia following anterior cruciate ligament (ACL) reconstruction. While there is significant data regarding the efficacy of specific types of regional blocks, there is less high-quality, clinical data informing the use of specific anesthetic agents.

Liposomal bupivacaine or Exparel (Pacira Biosciences) is a U.S. Food and Drug Administration approved agent, providing up to 72 hours of extended-release bupivacaine. This is promising given that the vast majority of opioids are consumed in the first 72 hours after ACL reconstruction. Exparel's use as a regional anesthetic has become increasingly common and has shown promise when utilized in ACL reconstruction but has not been studied after IPACK block. It is theorized, and has been substantiated in basic science literature, that the addition of dexamethasone to liposomal bupivacaine, may prolong the analgesic duration.

The objective of this study, was to characterize pain control and opioid consumption after the combination of ACB and IPACK regional anesthetic techniques with liposomal bupivacaine after ambulatory arthroscopic ACL reconstruction and to compare liposomal bupivacaine vs liposomal bupivacaine combined with dexamethasone to determine if the addition of dexamethasone significantly decreases postoperative narcotic use and prolongs analgesic effects in this same cohort.

Methods:

This is a prospective, blinded (patient, surgeon, research personnel) randomized controlled trial of a continuous series of primary bone-patellar tendon-bone or quadriceps tendon autograft ACL reconstructions at a single institution by 5 fellowship trained sports medicine surgeons. Patients were randomized in a 1:1 ratio into one of two arms:

Group #1 (Control): 30 mL Liposomal bupivacaine + 5 cc's of 0.5% bupivacaine evenly distributed and administered in adductor canal and IPACK block

Group# 2: 30 mL Liposomal bupivacaine + 10 mg of preservative free Dexamethasone 5 cc's of 0.5% bupivacaine evenly distributed and administered in adductor canal and IPACK block

All nerve infiltrations were performed by a single anesthesiologist. A standardized, postoperative multimodal pain control regimen was prescribed to all patients, which included scheduled ketorolac, acetaminophen, and metaxalone on a scheduled basis and 20 tablets of oxycodone (5 mg) for 'as needed' analgesia. Opioid consumption, visual analog scale (VAS) scores, pain control satisfaction, and subjective perceived block duration were recorded by patients via RedCap survey beginning on the day of surgery until postoperative day (POD) 7. Subjective perceived block duration was established by asking patients each day if they still felt the effects of their block (yes/no); the block duration was considered as the last day patients felt the effects of their block. T-tests were used to compare continuous data and Chi-Square tests were used to compare categorical data.

Results:

At the time of analysis, 91 patients underwent primary autograft ACL reconstruction with a ACB and IPACK regional anesthetics with liposomal bupivacaine and completed follow up surveys from the day of surgery through post-operation day 7. There were 42 patients in the group #1 and 49 patients in group #2. On average, patients consumed 1.6 oxycodone tablets [WJ1] from POD 0-7. There was no difference between group #1 and group #2 regarding the number of opioids consumed from POD 0-7 (1.2 tablets vs 1.9; $p = 0.4$). 38.7% of patients required no opioid analgesia postoperatively (41.7% vs. 37.5%; $p = 0.7$). The average VAS score was 36.1 (SD 17.9) with no differences between group #1 and group #2 (36.6 vs. 35.6; $p = 0.8$). The average subjective block duration was 5.33 days (SD 2.1). There was no difference in subjective block duration between group #1 and group #2 (5.1 vs. 5.5; $p = 0.4$). Overall, 85.4% of patients were satisfied with their pain control.

Conclusions:

Liposomal bupivacaine with or without dexamethasone via ACB and IPACK block provides excellent pain control and minimizes opioid consumption following autograft ACL reconstruction. Nearly 40% of patients did not require opioids postoperatively, making opioid-free ACL reconstruction a realistic possibility for many patients. When narcotics were required, the dose of opioids was minimal with patients requiring 1.6 tables of oxycodone on average. Interestingly, although prior reports suggest a prolonged duration when liposomal bupivacaine is combined with dexamethasone, we found no difference in opioid consumption, VAS score, patient satisfaction, or subjective block duration between patients who did and did not receive dexamethasone with their local block .